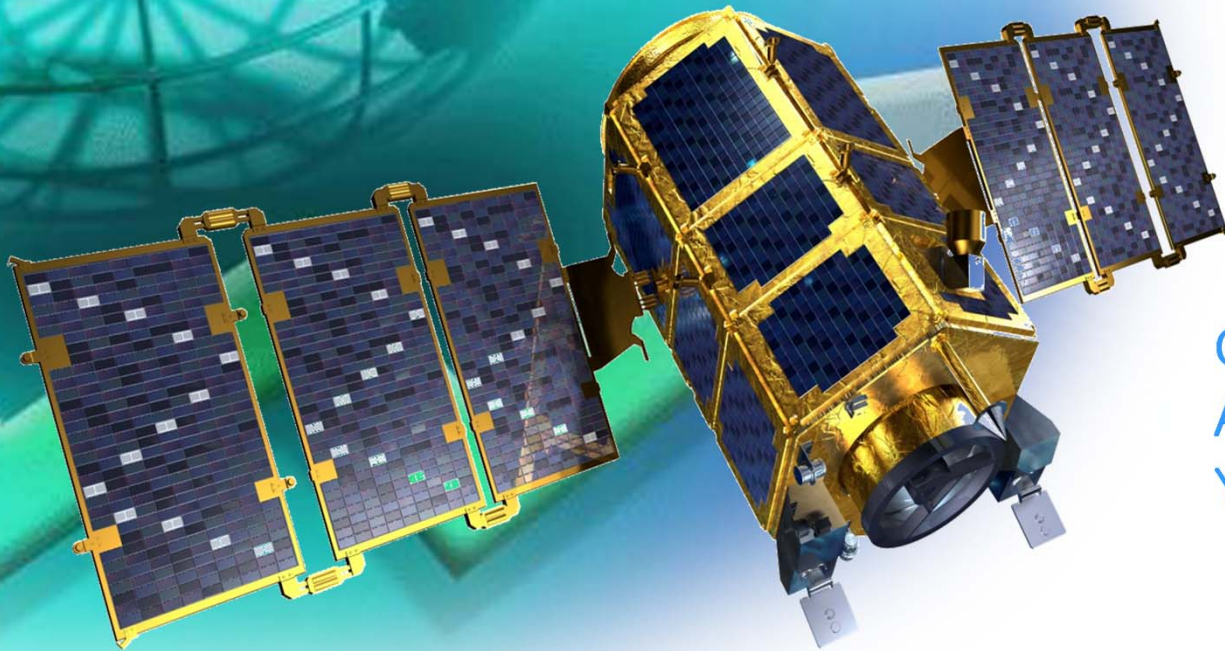
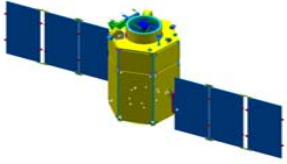


Fall 2011

Spacecraft Systems



Chandeok Park
Astrodynamics & Control Lab.
Yonsei University

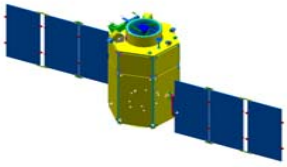


Chapter 1

SPACE MISSION ANALYSIS & DESIGN PROCESS

1.1 Introduction & Overview

1.2 The Space Mission Life Cycle



◆ Space Mission Analysis and Design

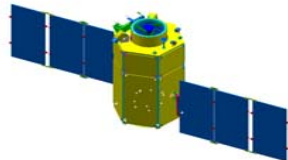
- begins with one or more broad objectives and constraints
- then proceeds to define a space system that will meet them at the lowest possible cost.

◆ To get the most performance for the money spent

- we must require of the system only it can reasonably achieve.

◆ Analysis and Design are

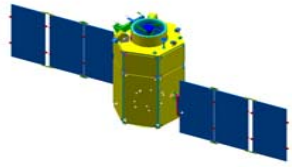
- iterative
- gradually refining both the requirements and methods of achieving them.



1.1 Introduction & Overview

TABLE 1-1. The Space Mission Analysis and Design (SMAD) Process. Tables of this type appear throughout the book. The far right column refers to sections in the book that give details of each step. See text for further explanation.

Typical Flow	Step	Section
	Define Objectives { 1. Define broad objectives and constraints 2. Estimate quantitative mission needs and requirements	1.3 1.4
	Characterize the Mission { 3. Define alternative mission concepts 4. Define alternative mission architectures 5. Identify system drivers for each 6. Characterize mission concepts and architectures	2.1 2.2 2.3 2.4
	Evaluate the Mission { 7. Identify critical requirements 8. Evaluate mission utility 9. Define mission concept (baseline)	3.1 3.3 3.4
	Define Requirements { 10. Define system requirements 11. Allocate requirements to system elements	4.1 4.2–4.4



1.1 Introduction & Overview

◆ MISSION STATEMENT

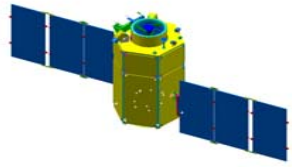
- Ex) Figure 1–1 FireSat

◆ Step 1: Define Broad Objectives and Constraints

- We define what the mission needs to achieve.
- What are our qualitative goals and why?
- Should come largely from Mission Statement

◆ Step 2: Estimate Quantitative Mission Needs and Requirements

- Quantifies how well we wish to achieve the broad objectives, given our needs, applicable technology, and cost constraints.
 - Ex) FireSat with 100m positioning accuracy
- Should be subject to trade as we go along



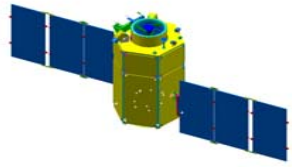
1.1 Introduction & Overview

◆ Step 3: Define Alternative Mission Concepts

- A Mission Concept (or Concept of Operations) is a broad statement of how the mission will work in practice:
 - low-cost approach, different orbits, different wavelength bands, etc.

◆ Step 4: Define Alternative Mission Architectures

- It defines alternate combinations of mission elements or the space mission architecture to meet the requirements of the mission concept.
- The space mission architecture = the mission concept + a definition of each of the elements of the mission.



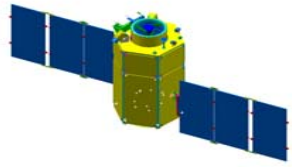
1.1 Introduction & Overview

◆ Step 5: Identity System Drivers for Each

- We identify the principal cost and performance drivers for each alternative mission concept
 - number of satellites, altitude, power, instruments size and weight.

◆ Step 6: Characterize Mission Concepts & Architectures

- It defines in detail what the system is and does.
- We determine the power, weight, pointing budgets,
- We decide what to process on the ground or in space.



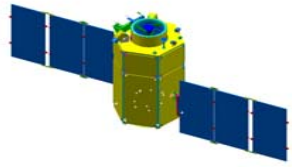
1.1 Introduction & Overview

◆ Step 7: Identity Critical Requirements

- We return to our initial quantitative requirements and identify the critical requirements (responsible for determining the cost and complexity of the system)
 - Ex) altitude, coverage, aperture, resolution etc.

◆ Step 8: Evaluate Mission Utility

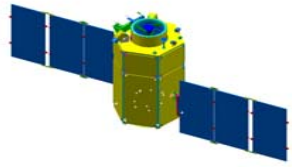
- We quantify how well we are meeting both the requirements and broad objectives as a function of either cost or key system–design choices.
- We provide the decision maker a single chart of potential performance vs. cost.
 - the percent of fires detected within 2 hours vs. the aperture of the instrument
 - the delay time in detecting fires vs. altitude and number of satellites



1.1 Introduction & Overview

◆ Step 9: Define Mission Concept (Baseline)

- We select one or more baseline system designs.
- A baseline design is a single consistent definition of the system which meets most or all of the mission objectives.
 - Ex) resolution and coverage rates which correspond to the assigned altitude, aperture, s/c weight etc.
- It is much more feasible to look at the impact of varying each of elements individually relative to one or two baseline designs.



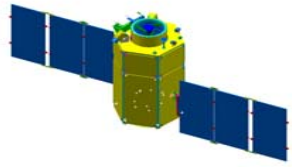
1.1 Introduction & Overview

◆ Step 10: Define System Requirements

- We translate the broad objectives and constraints of the mission into well-defined system requirements.

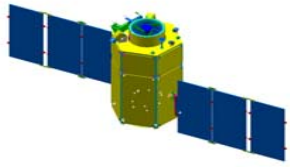
◆ Step 11: Allocate Requirements to System Elements

- We flow down or allocate numerical requirements to the components of the overall space mission in the same way that a budget allocates weight and power to the spacecraft's components.



1.1 Introduction & Overview

- ◆ Analysis & Design Process for Space Missions is evolving
- ◆ Anticipation
 - Technological limits are continually giving way to political & economical Limits.
 - Increasing use of onboard processing
 - Has been conservative in this direction
 - To pursue more autonomy
 - Continual emphasis on low-cost missions
 - Small/Micro Satellites & Their Constellations
 - Multi-mission Concept/Architectures or Integration of Operational Aspects of Different Missions



1.2 The Space Mission Life Cycle

◆ Concept Exploration

- The initial study phase, which results in a broad definition of the space mission and its components.

◆ Detailed Development

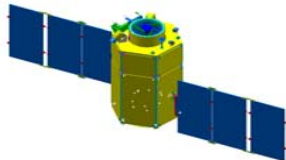
- The formal design phase, which results in a detailed definition of the system components and, in larger programs, development of test hardware or software.

◆ Production and Deployment

- The construction of the ground and flight hardware and software and launch of the satellite.

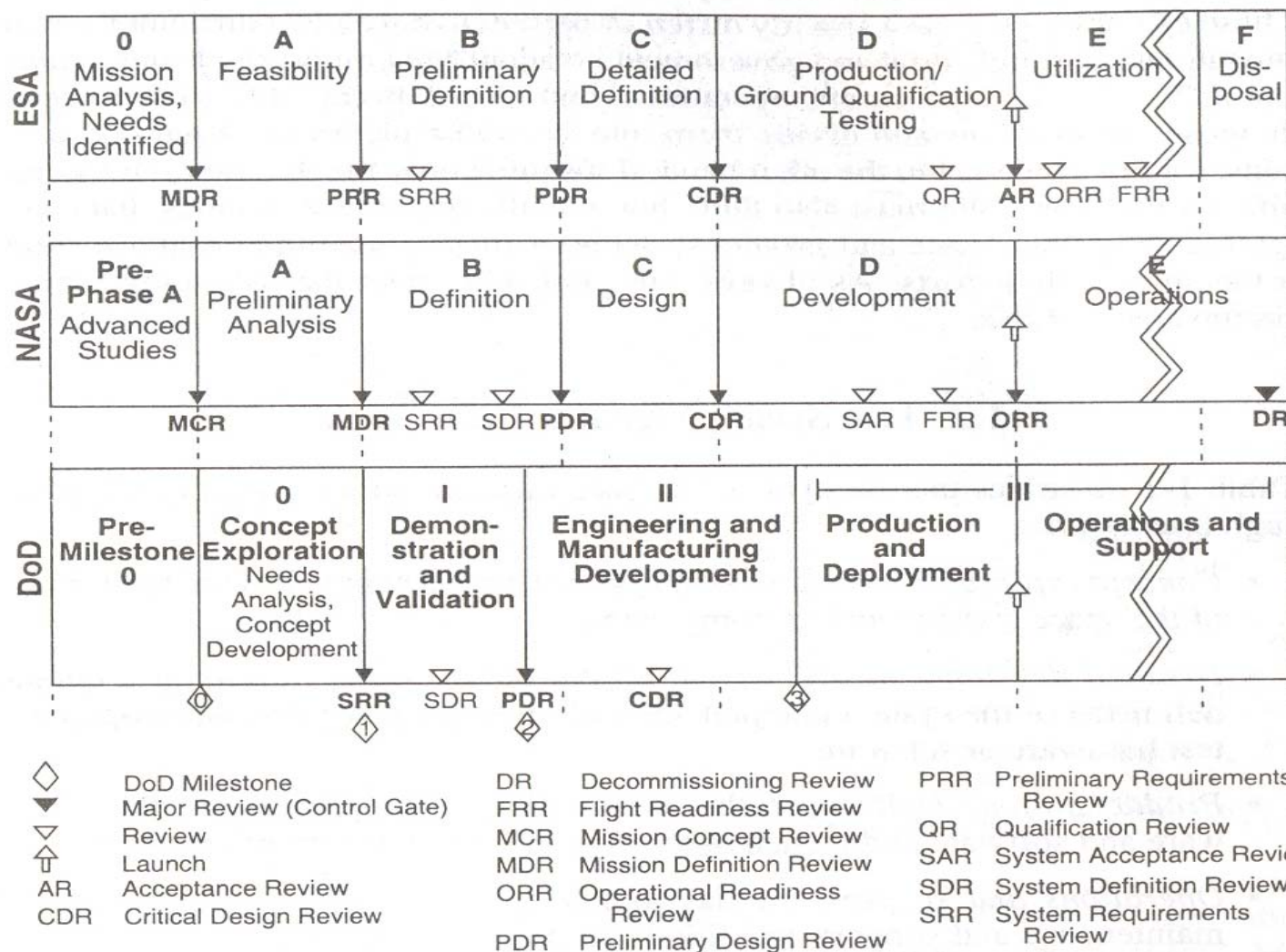
◆ Operation and Support

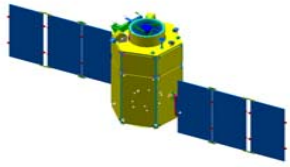
- The day-to-day operation of the space system, its maintenance and support, and finally its deorbit or recovery at the end of the mission life



1.2 The Space Mission Life Cycle

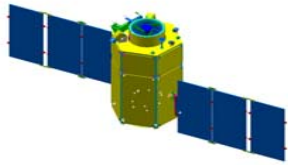
TABLE 1-2. Space Program Development Phases. Every space program progresses through the top-level phases. Subphases may or may not be part of a given program. The time required to complete the process varies with the scope. (Diagram courtesy R. Bertrand.)





1.2 The Space Mission Life Cycle

- ◆ Key players for procurement and operating policies and procedures
 - Space Mission Operator
 - control and maintain the space and ground assets
 - End User or Customer
 - receive and use the products and capability of the space mission.
 - Developer
 - is the procuring agent.
 - Sponsor
 - performs long range planning
 - and estimates budgetary needs and available funding.



1.2 The Space Mission Life Cycle

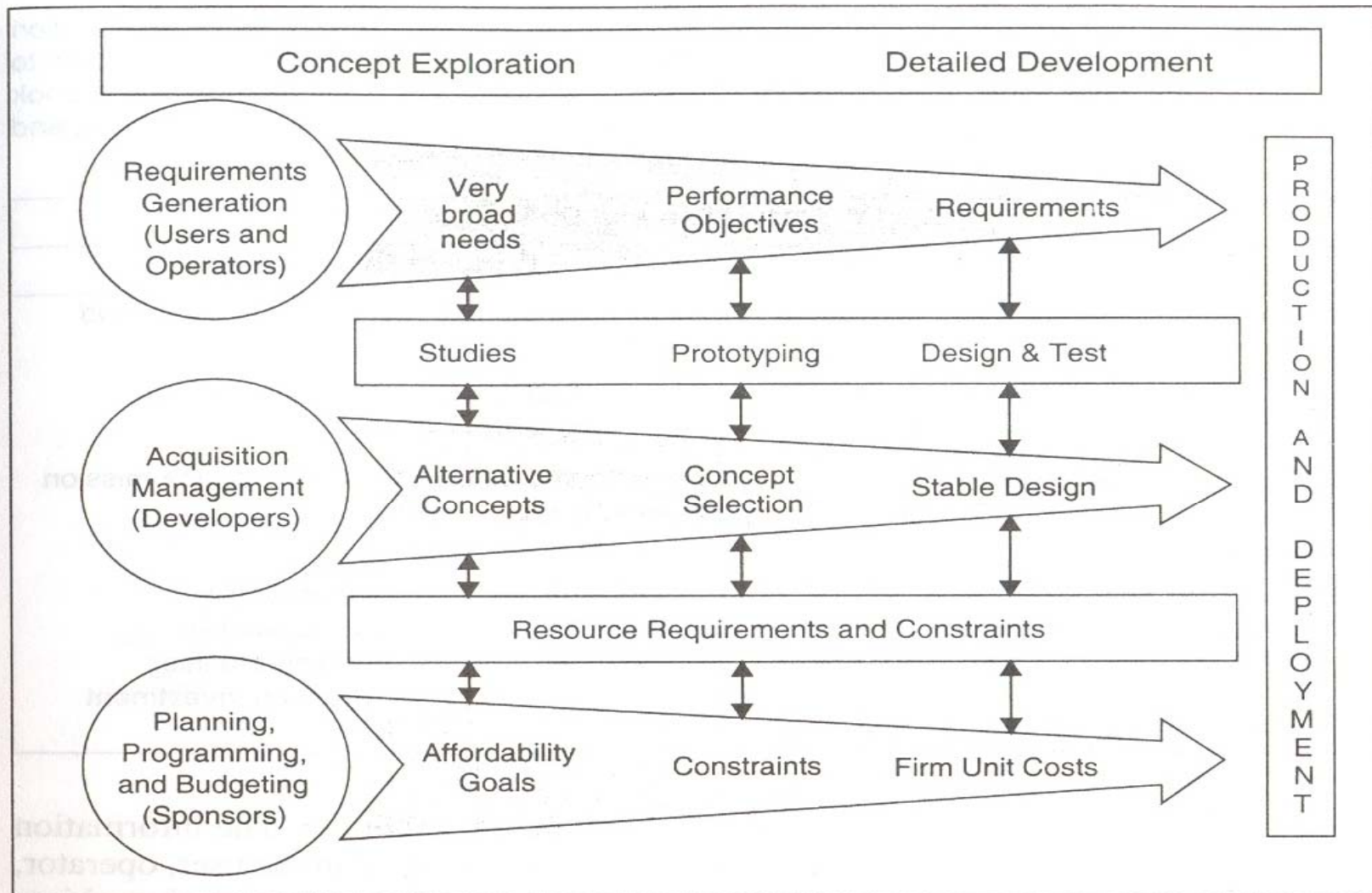
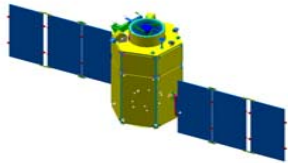


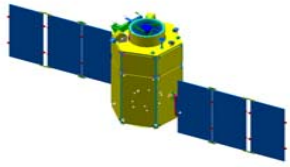
Fig. 1-2. Organizations and People that Play Key Roles in Space Missions. During concept exploration users and operators provide requirements, developers create the design of the mission and systems, while sponsors provide funding.



1.2 The Space Mission Life Cycle

TABLE 1-3. Further Breakdown of Concept Exploration Phase. During concept exploration the operator and end users define their needs and requirements and pass them to the developing organization for concept development. A basic premise of this book is that the operator, user, and developer should work together to create realistic and affordable mission objectives and requirements that meet user needs.

Concept Exploration and Definition		
Needs Analysis	Concept Development	
Generate potential requirements based on Mission objectives Concept of operations Schedule Life-cycle cost and affordability Changing marketplace Research needs National space policy Long-range plan for space Changing threats to national defense Military doctrine New technology developments Commercial objectives	Reassess potential requirements generated during needs analysis Develop and assess alternative mission operations concepts Develop and assess alternative space mission architectures Estimate	performance life-cycle cost schedule risk
		supportability produceability funding profiles return on investment



1.2 The Space Mission Life Cycle

- ◆ To explore a concept successfully,
 - We must remove the walls between the sponsor, space operator, users, and developer
 - And become a TEAM.

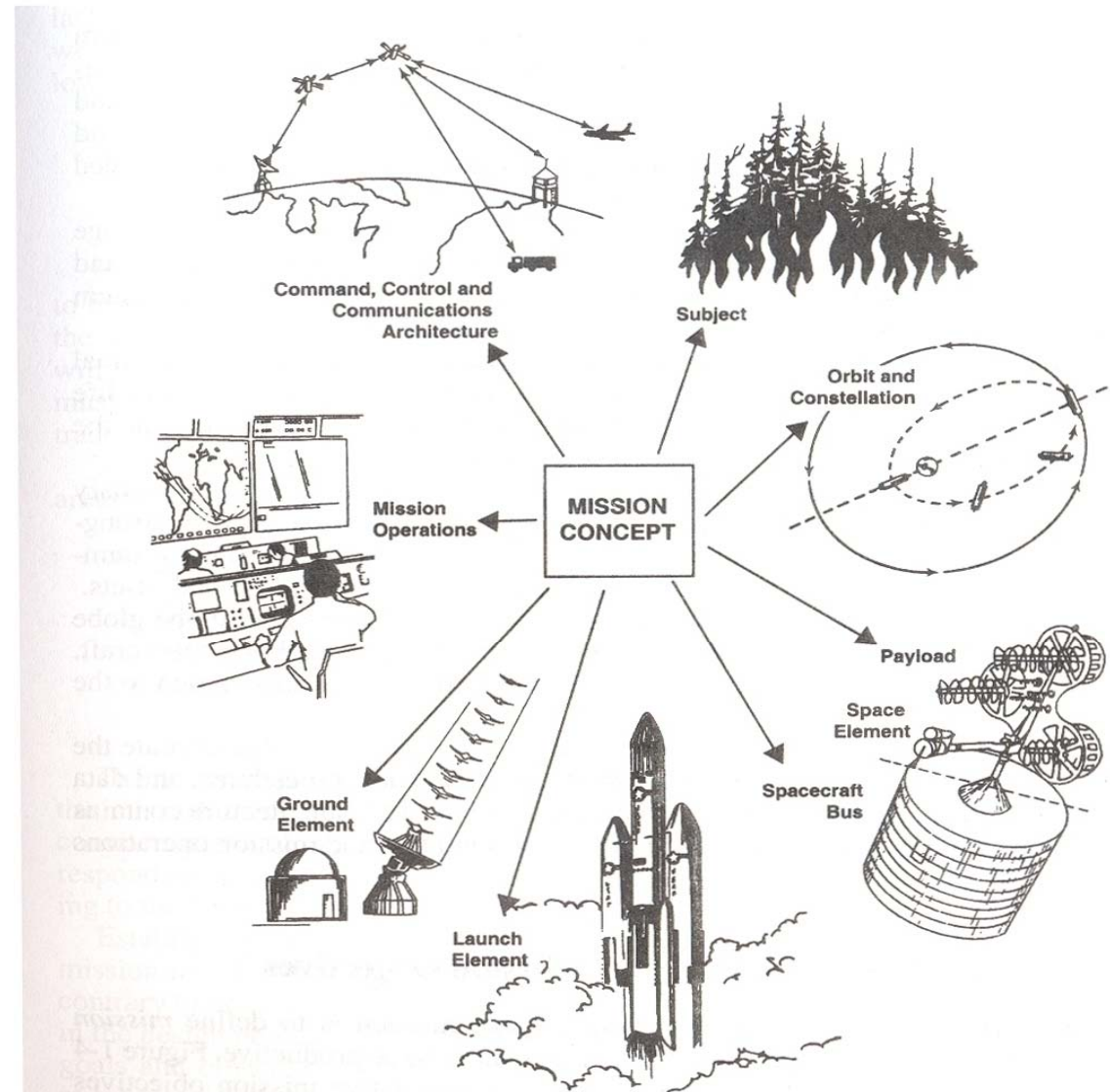


Fig. 1-3. Space Mission Architecture. All space missions include these basic elements to some degree. See text for definitions. Requirements for the system flow from the operator, end user, and developer and are allocated to the various mission elements.